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Multiple Linear Regression Model of Environmental Variables, Predictors of Global Solar Radiation in the Area of East Lima, Peru

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Abstract. Multiple regression models are very relevant to predict values using predictor variables. The objective of this study was to predict the global solar radiation in the year 2019 in the area of East Lima, Peru. Three continuous quantitative predictor variables were analyzed: temperature, humidity, wind speed and the response variable was global solar radiation, resulting in a model with excellent significance $p<0.001$ that shows the prediction is effective. The multiple linear regression method was used, finding an average global radiation of 175 W/m2 and predictor variables with average temperature of 19.2 °C, humidity 23.9% and wind speed 1.77 m/s, with the highest temperature in summer recorded at 24.6°C, the highest humidity of 51.2% in autumn, the highest wind speed in summer at 2.63 m/s and the highest maximum global solar radiation in spring with 183 W/m^2.

1. Introduction

In the eastern area of Lima there is a meteorological station, on the campus of the Universidad Peruana Unión (UPeU), which allows us to monitor the environmental variables of the sector. The purpose of this study was to evaluate the predictive variables of solar radiation using the multiple regression model. [1] state that "Distribution modeling and mapping was performed to identify key environmental variables and predict the potential range of the species under current climatic conditions in a study conducted in India". They accessed species occurrence data from primary and secondary sources, which they used after cleaning and eliminating bias. Climatic, topographic, cover and soil variables of 1 km resolution were used as environmental predictors.

Predicting a variable from modeling environmental variables by multiple linear regression has been applied in other studies [2], [3] . Modeling is an important step when we want to understand the complexity of environmental behavior and predict changes in the environment. [4] studied the synergy between solar radiation and environmental variables such as temperature. They found that the main factor affecting the deterioration of PCM properties is low temperature. The relationship of solar radiation and temperature was evaluated by [5] in China and [6] in Junín.

The environmental variables in the study are temperature, humidity, wind speed and global solar radiation. Temperature and humidity are very relevant environmental variables in environmental characterization; [7] considered them in a recent study on population, temperature and absolute humidity in settings such as China, England, Germany and Japan. Likewise [8] implemented an intelligent temperature and humidity monitoring system for remote environmental monitoring. [9]

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evaluated the accuracy of linear regression equations based on meteorological data such as minimum and maximum temperature, relative humidity, wind speed, solar radiation, extraterrestrial radiation and vapor pressure deficit. In this study we estimate global solar radiation based on environmental variables (temperature, humidity, wind speed) by applying multiple linear regression analysis.

[10] mention in their study on the use of maximum and minimum temperature which estimates solar radiation by equation (1) represented by:

$$\frac{R_s}{R_a} = K_r \left(T_{\max} - T_{\min}\right)^{0.5} \tag{1}$$

Where Kr is an empirical coefficient that was recommended to be 0.16 for inland regions and 0.19 for coastal regions, associated to a linear regression model described by equation (2).:

$$\frac{R_s}{R_a} = -0.255 + 0.246\left(T_{\max} - T_{\min}\right)^{0.5} \tag{2}$$

Where Kr is determined locally.

2. Material and methods

This study used a group of 19 500 records of measurements made every 5 minutes, 24 hours a day, in the year 2019, measuring meteorological variables such as global ambient temperature, output humidity and wind speed as predictor variables, and global solar radiation as the dependent one [11] . The instrument used for the measurement of meteorological variable data was the Vantage Pro2 sensor [12]. For this study, the multiple regression method was used to generate the predictive model of global solar radiation suitable for the eastern area of Lima. The analysis was performed by season for the year 2019 (winter, autumn, spring and summer) and efficient results were obtained from the model. [13] mentions the usefulness of the applied method and how it contributed to model problems related to meteorological variables in the area of East Lima, which will be beneficial in exploring the interconnection between the environmental setting, the climate of the earth's atmosphere and what may happen in the future as the climate is monitored, by monitoring meteorological variables.

The predictions of solar radiation have a non-stationary behavior due to multiple parameters and atmospheric conditions that include the large amount and scale of the data involved [14] , which is why this study presents a predictive model based on multiple regression considering meteorological variables that have a great incidence in the global solar radiation is presented, thus considering one of the models that explain the reality in the area of East Lima, Peru.

The normality of the variables under study was tested, using the Kolgomorov Smirnov normality test, as well as homoscedasticity, linearity, independence and non-collinearity, validating the multiple regression assumptions [15]. The relationships between variables were analyzed using simple correlation, yielding a significant relationship between the predictor variables and the response variable [16].

The multiple regression analysis was applied as a means to evaluate statistically significant variables of the measurements made with the Vantage Pro2 instrument during 2019, in which significant descriptive statistics were obtained in the study variables [17].

Figure 1 shows the location of the Vantage Pro2 instrument located at the Universidad Peruana Union campus in Lima Peru that measured the data of the meteorological variables under study during 2019 [18].

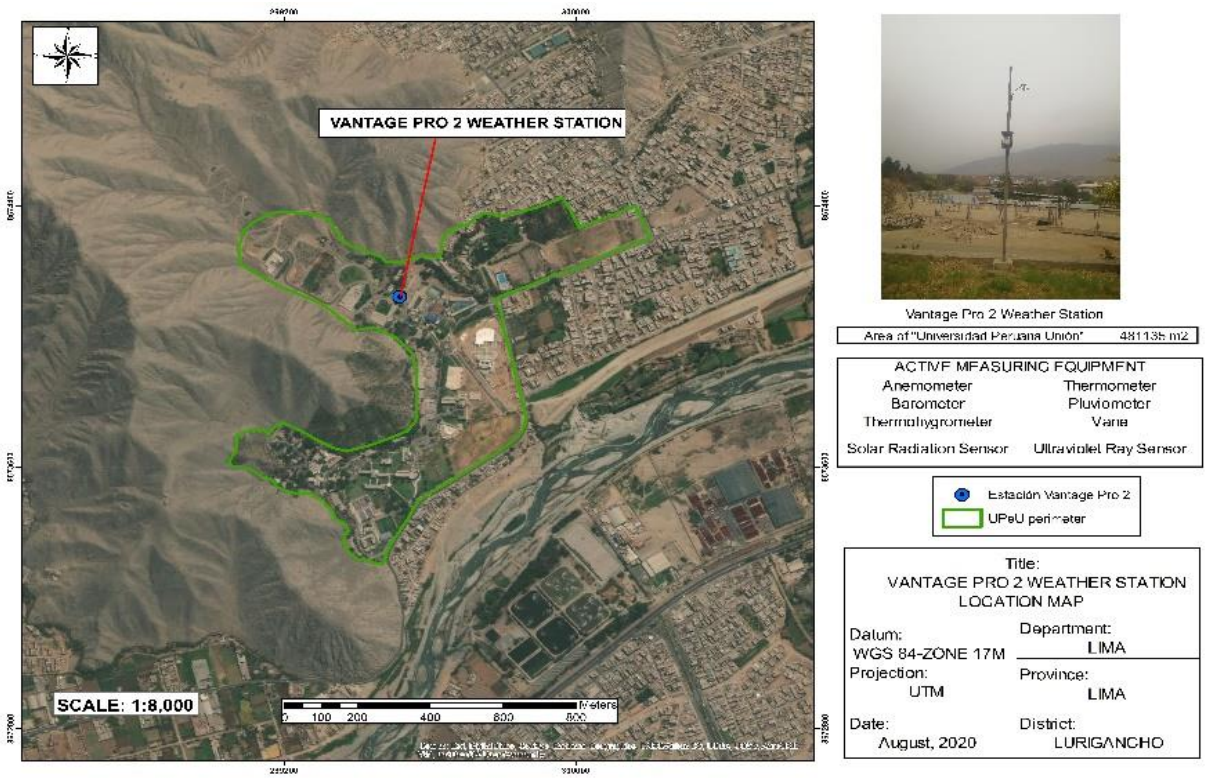

Figure 1. Location of the meteorological station in the East zone, Lima.

3. Results

The descriptive statistics by environmental variable are described in Table 1, which shows an average temperature of 19.2°C, an average humidity of 23.9%, an average wind speed of 1.77 m/s and 175 W/m^2 of global solar radiation, contrasting with the average obtained in the Mantaro Valley with 175.7 W/m^2 studied by [19], there is a mean difference of 0.7 W/m^2.

Table 1. Descriptive statistics of the study variables.

	Temperature(°C)	Humidity (%)	Wind speed (m/s)	Global solar radiation (W/m^2)
N	19551	19532	19564	19564
Missing	13	32	0	0
Mean	19.2	23.9	1.77	175
Median	19.1	20.0	0.00	0.00
Minimun	11.9	0.00	0.00	0
Maximun	29.8	60	16.1	1062

The correlation matrix of the variables is shown in Table 2, which indicates that there is a very good correlation of the predictor variables with the response variable, fulfilling the assumption of the model.

Table 2. Correlation matrix of the study variables.

		Temperature (°C)	Humidity (%)	Wind speed (m/s)	Global solar radiation (W/m^2)
Temperature(°C)	Pearson's r p-value	--- 			
Humidity (%)	Pearson's r p-value	0.859 <0.001	---		
Wind speed (m/s)	Pearson's r p-value	0.570 <0.001	0.336 <0.001	---	
Global solar radiation (W/m^2)	Pearson's r p-value	0.454 <0.001	0.182 <0.001	0.708 <0.001	---

The environmental variables analyzed recorded an average temperature of 19.2 °C with a minimum of 11.9 °C and a maximum of 29.8 °C; likewise the humidity is presented with an average of 23.9% with a minimum of 0% and a maximum of 60%; also the wind speed with an average of 1.77 m/s with a minimum of 0 m/s and a maximum of 16.1 m/s and solar radiation with an average of 175 W/m^2 with a minimum of 0 and a maximum of 1062 W/m^2. The measurements of the variables by season in the year 2019: in winter an average temperature (15.9 °C), Humidity (6.32%), wind speed (1.28 m/s), solar radiation (164 W/m^2) was recorded; likewise in Autumn an average temperature (22.6 °C), Humidity (51.2%), wind speed (2. 53 m/s), solar radiation (180 W/m^2); also in spring an average temperature (17.3 °C), humidity (9.87%), wind speed (1.27 m/s), solar radiation (183 W/m^2) was recorded and finally in summer an average temperature (24.6 °C), humidity (50.2%), wind speed (2.63 m/s), solar radiation (180 W/m^2) was recorded.

Table 3. Model Coefficients for Global Solar Radiation

			Significance	
Predictor	Estimate	SE	t	p
Intercept	-394.55	10.260	-38.5	< 0.001
Temperature	31.12	0.674	46.2	< 0.001
Humidity	-5.01	0.109	-46.2	< 0.001
Wind speed	52.06	0.577	90.2	< 0.001

In table 3 we observe the coefficients of the multiple regression model with intercept at -394.55 ($p<0.001$), a temperature coefficient of 31.12 ($p<0.001$), a humidity coefficient of -5.01 ($p<0.001$) and a wind speed coefficient with a value of 52.06 ($p<0.001$), determining the confinable multiple regression model represented in equation (3)

$$y = -394.55 + 31.12x_1 - 5.01x_2 + 52.06x_3 \quad (3)$$

The multiple regression model indicates that for every one unit increase in temperature, global solar radiation will increase by 31.12 W/m^2, likewise for every one unit increase in humidity, solar radiation will decrease by 5.01 W/m^2, and for every one unit increase in wind speed, solar radiation will increase by 52.06 W/m^2.

The 95% confidence intervals of the multiple regression model, the intercept (-394.55) had a variation between -414.66 and -374.44 with a standard deviation of 10.260, the temperature (31.12) a variation between 29.80 and 32. 44 with a standard deviation of 0.674, humidity (-5.01) a variation between -

5.23 and -4.80 with a standard deviation of 0.109 and wind speed (52.06) a variation between 50.93 and 53.19 with a standard deviation of 0.577.

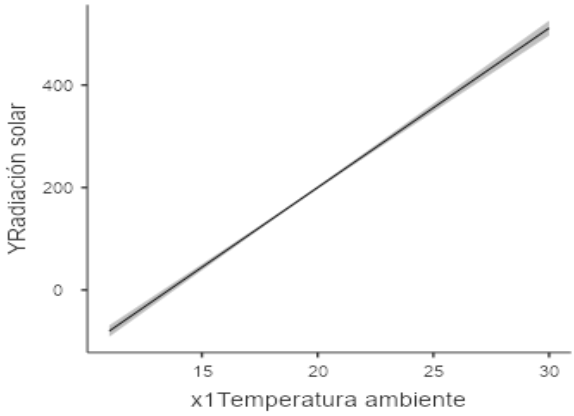

Figure 2. Marginal mean temperature

Figure 2 shows the estimate of the population marginal mean for the temperature predictor variable (19.2 °C), with a 95% confidence interval ranging from 14.8 to 23.6 with a standard deviation of 1.23, which is positively related to global solar radiation.

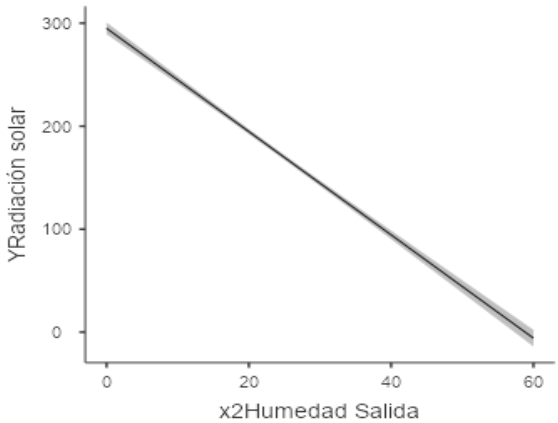

Figure 3. Marginal mean humidity

Figure 3 shows the estimate of the population marginal mean for the humidity predictor variable (23.9242), with a 95% confidence interval ranging from 0.0299 to 47.8186 with a standard deviation of 1.23, which is inversely related to global solar radiation.

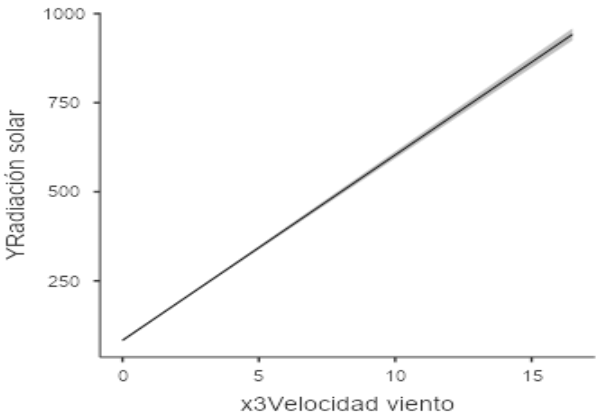

Figure 4. Marginal mean for wind speed

Figure 4 shows the estimate of the population marginal mean for the wind speed predictor variable (1.77), with a 95% confidence interval ranging from -1.03 to 4.57 with a standard deviation of 1.23, which is positively related to global solar radiation.
Figure 5 shows the graph of the residuals of the solar radiation response variable, which shows predictive linearity with a very good level of accuracy (p_value<0.001)..

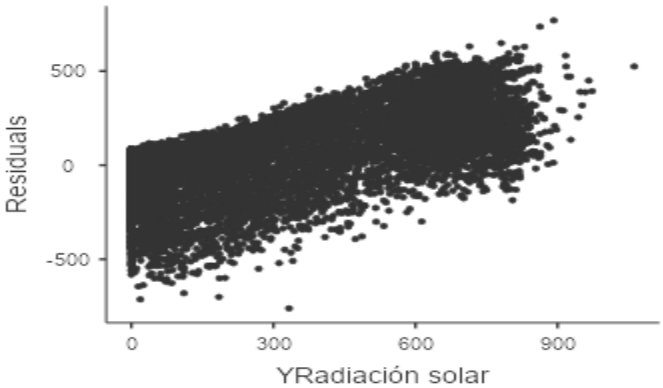

Figure 5. Residuals of solar radiation

Table 4. Model Fit Measures

Model	R	R^2
1	0.746	0.556

In Table 4, we observe that the coefficient of determination R2 of the model explains 55.6% of the original uncertainty and a correlation coefficient of 0.746, which is significant [20].

Table 5. Results of the ANOVA.

Results	Sum of Squares	Mean Square	Significance F	Significance p
Temperature	6.34e+7	6.34e+7	2131	< 0.001
Humidity	6.35e+7	6.35e+7	2132	< 0.001
Wind Speed	2.42e+8	2.42e+8	8142	< 0.001
Residuals	5.81e+8			

In table 5, we observe that all the environmental variables under study have a significance lower than the alpha level α=0.05 (p<0.001) so the multiple regression model is validated at a confidence level of 95%.

Table 6. Sample size and averages by stations

Season	Temperature (Mean)	Humidity (Mean)	Wind speed (Mean)
Winter	6676 (15.9)	6668 (6.32)	6681 (1.28)
Fall	3178 (22.6)	3179 (51.2)	3179 (5.53)
Spring	5564 (17.3)	5551 (9.87)	5570 (1.27)
Summer	4133 (24.6)	4134 (50.2)	4134 (2.63)

Table 6 shows the 19551 records made of the meteorological variables by station in the year 2019 with the Vantage Pro instrument, determining the averages by seaon in its three predictor variables.

4. Discussion

In the study by [10], they found a regression model, based on Hargreaves equation, which estimates solar radiation. Their model is similar with the one found in the present study in its prediction and accuracy in calculating global solar radiation.

The study by [11] mentions that correlation in their regression model generated by the Dumas equation has a correlation coefficient located in the interval 0.8< r <0.9 which shows a good performance of their model. In the environmental predictor variables and global solar radiation, they had a correlation of 0.746 which makes the multivariate linear regression model confinable.

The study by [6] classifies the temperature by seasons of the year, finding that the maximum and minimum temperatures decrease in the months of May, June and July, which applied the Bristow model to estimate the daily, monthly and seasonal solar radiation. Applying the multiple regression model, a temperature of 24.6 °C was found in January, a humidity of 51.2 % in March, a wind speed of 2.63 m/s in January and a global solar radiation of 183 W/m^2 in September 2019.

Regarding the average global solar radiation of the study is 175 W/m^2, in which there is a similarity with the findings found by [19] with a value of 175.7 W/m^2 which makes a mean difference of 0.7 W/m^2.

5. Conclusion

According to the objective of this study, a confinable multivariate regression model ($p<0.001$) was found that allows for the prediction of global solar radiation (175 W/m^2) in the East Lima region based on the environmental predictor variables on average; temperature (19.2°C), humidity (23.9%) and wind speed (1.77 m/s). Seasonal averages were also obtained: the highest temperature was summer with 24.6°C, autumn with the highest humidity of 51.2%, summer with a wind speed greater than 2.63 m/s and spring with a maximum global solar radiation of 183 W/m^2.

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